Arista Access Switch (WFD-ACC-31) - QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Troubleshooting and Resolution SOP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1093

Description

The failure domain of QoS congestion events on the Arista access switch (WFD-ACC-31) at the Whitefield POP is primarily attributed to the interplay between the switch's hardware capabilities, the configured Quality of Service (QoS) policies, and the underlying network traffic patterns. The Arista 7000 series switches employ a distributed architecture, where each line card is responsible for forwarding packets independently. This design, while increasing overall performance and reducing latency, introduces potential bottlenecks in specific scenarios, such as microbursts or interface tail drops. Furthermore, the QoS policies implemented on the switch, including Class-Based QoS (CBQoS) and queueing disciplines like Priority Queue (PQ) and Hierarchical PQ (HPQ), play a crucial role in managing network traffic and mitigating congestion. However, misconfigurations, inadequate policy implementation, or insufficient hardware resources can lead to QoS queue starvation, exacerbating congestion and resulting in the ALARM_1093 event. To resolve this issue, it is essential to identify the root cause, assess the network traffic patterns, and adjust the QoS policies and switch configurations accordingly.

Pre-requisite Checks

- Verify that the Whitefield POP's remote hands are dispatched and available for on-site assistance.

- Check the status of the redundant link connecting the WFD-ACC-31 to the upstream network device.

- Ensure that the network traffic patterns during the failure event are documented and available for analysis.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to queue starvation	High
Inadequate hardware resources (e.g., insufficient buffer pool size)	Medium
Malfunctioning or faulty hardware components (e.g., line cards or interfaces)	Low

Diagnostic Commands & Expected Outputs

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Command: show hardware resource usage, Expected Output: Buffer pool utilization, memory usage, and other hardware resource metrics,
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Description: This command helps identify potential hardware resource bottlenecks, such as insufficient buffer pool size, that may contribute to QoS congestion events.

Command: show qos policy, Expected Output: Configured QoS policies, including class maps, policy maps, and queueing disciplines, Description: This command allows for the verification of existing QoS policies, including their configuration, class maps, policy maps, and queueing disciplines.

Command: show interface , Expected Output: Interface statistics, including packets, bytes, and errors, Description: This command helps identify potential interface-related issues, such as microbursts or tail drops, that may be contributing to QoS congestion events.
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Resolution Steps

Step 1: Step: Adjust QoS policies to prevent queue starvation, Command: config t; policy-map class bandwidth 100, Description: This command adjusts the QoS policy to prevent queue starvation by allocating sufficient bandwidth to the affected class.

Step 2: Step: Increase buffer pool size to address hardware resource bottlenecks, Command: config t; hardware resource buffer-pool-size 1024, Description: This command increases the buffer pool size to address potential hardware resource bottlenecks, reducing the likelihood of QoS congestion events.

Step 3: Step: Replace faulty or malfunctioning hardware components, Command: config t; hardware component replace, Description: This command initiates the replacement of faulty or malfunctioning hardware components, such as line cards or interfaces, to prevent future QoS congestion events.

Post-Check Verification

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Command: show hardware resource usage, Expected Output: Buffer pool utilization, memory usage, and other hardware resource metrics within acceptable ranges, Description: This command verifies that the hardware resource usage is within acceptable ranges, ensuring that the QoS congestion event has been resolved.

Command: show qos policy, Expected Output: Configured QoS policies, including class maps, policy maps, and queueing disciplines, with adjustments made to prevent queue starvation, Description: This command verifies that the QoS policies have been adjusted to prevent queue starvation, ensuring that the QoS congestion event has been resolved.

Command: show interface , Expected Output: Interface statistics, including packets, bytes, and errors, within acceptable ranges, Description: This command verifies that the interface-related issues, such as microbursts or tail drops, have been addressed, ensuring that the QoS congestion event has been resolved.
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Roll-back Procedures

- Step: Restore original QoS policies, Command: config t; policy-map class bandwidth , Description: This command restores the original QoS policies, including the bandwidth allocation, to the affected class.

- Step: Revert buffer pool size changes, Command: config t; hardware resource buffer-pool-size , Description: This command reverts the buffer pool size changes made to address hardware resource bottlenecks, returning to the original configuration.

- Step: Revert hardware component replacement, Command: config t; hardware component revert, Description: This command reverts the hardware component replacement, restoring the original component.